Revision Comparison

A grammar edit beside a multiline equation

Task. Correct subject–verb agreement. Preserve the setup, parameter dimensions, regularization condition, equation, and concluding statement. This is synthetic material, not an experimental or theorem-validity report.

Versions. `before.tex` and `after.tex`; source hashes are recorded in `identity.json`. Only C1 changes the source text.

C1: original passage

Location. Objective-introduction sentence, anchor "We minimizes". The surrounding setup and equation are included to show the edit's scope.

Let $X \in \mathbb{R}^{n \times d}$, $y \in \mathbb{R}^n$, $n \geq 1$, and $\lambda > 0$. The parameter is $\theta \in \mathbb{R}^d$.

We minimizes the following regularized objective:

$$\begin{aligned} F(\theta) &= \frac{1}{2n} \sum_{i=1}^{n} \left(\sum_{j=1}^{d} X_{ij}\theta_j - y_i \right)^2 + \frac{\lambda}{2} \|\theta\|_2^2 \\ &= \frac{1}{2n} \|X\theta - y\|_2^2 + \frac{\lambda}{2} \sum_{j=1}^{d} \theta_j^2. \end{aligned} \tag{B1}$$

The two lines of Equation (B1) express the same objective.

C1: revised passage

Location. The same sentence, anchor "We minimize".

Let $X \in \mathbb{R}^{n \times d}$, $y \in \mathbb{R}^n$, $n \geq 1$, and $\lambda > 0$. The parameter is $\theta \in \mathbb{R}^d$.

We minimize the following regularized objective:

$$F(\theta) = \frac{1}{2n} \sum_{i=1}^{n} \left(\sum_{j=1}^{d} X_{ij} \theta_j - y_i \right)^2 + \frac{\lambda}{2} \|\theta\|_2^2 \tag{A1}$$
$$= \frac{1}{2n} \|X\theta - y\|_2^2 + \frac{\lambda}{2} \sum_{j=1}^{d} \theta_j^2.$$

The two lines of Equation (A1) express the same objective.

Revision note: Editor-proposed: correct the verb agreement with "We." The subject, action, and mathematical content are preserved.

Meaning and review. Exposition-only. The verb ending is the only change between the source fragments. No proof or experimental claim has been audited. Formulas remain full-size and are not enclosed in inline color macros.

Presentation mapping. The source label `eq:objective` is displayed as `before:eq:objective` (B1) and `after:eq:objective` (A1). Counters are reset separately. These labels belong to the comparison wrapper; the source files are unchanged.

Legend. Purple marks a rewritten word; the source quotations include the same word replacement. IDs and side headings identify the change without relying on color. With annotations off, the two source versions remain visible.